# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Stainless steel contains iron with: 1

**(a)** zinc and copper **(b)** nickel and chromium
**(c)** carbon only **(d)** tin and lead

**2.** Aqua regia is a mixture of: 1

**(a)** conc. HCl and conc. $H_2SO_4$
**(b)** conc. HCl and conc. $HNO_3$ in 3 : 1 ratio
**(c)** dilute HCl and dilute $HNO_3$
**(d)** conc. $HNO_3$ and conc. HCl in 3 : 1

**3.** Roasting is the process of heating ore: 1

**(a)** with water **(b)** with carbon
**(c)** in excess of air **(d)** in the absence of air

**4.** Galvanisation is coating iron with: 1

**(a)** zinc **(b)** copper **(c)** tin **(d)** nickel

**5.** The most ductile metal is: 1

**(a)** iron **(b)** gold **(c)** zinc **(d)** lead

**6.** Calcination is heating of ore: 1

**(a)** in excess air **(b)** in hydrogen
**(c)** in limited or no air **(d)** with water

**7.** The surface of aluminium is protected from corrosion by a layer of: 1

**(a)** aluminium sulphide **(b)** aluminium hydroxide
**(c)** aluminium oxide **(d)** aluminium chloride

**8.** Which of the following is the best conductor of electricity? 1

**(a)** silver **(b)** iron **(c)** mercury **(d)** lead

**9.** The thermit reaction is used to: 1

**(a)** weld railway tracks **(b)** extract sodium
**(c)** make glass **(d)** prepare soap

**10.** Which of the following is a non-metal and shines? 1

**(a)** carbon (coal) **(b)** phosphorus **(c)** sulphur **(d)** iodine

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** What are alloys? Give two examples. 2

**12.** Name two metals which will displace hydrogen from dilute acids and two which will not. 2

**13.** State two physical properties of ionic compounds. 2

**14.** Why is aluminium considered a reactive metal yet used for cooking utensils? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Why do ionic compounds have high melting points? Why do they not conduct in solid state? 3

**16.** Name three methods of preventing rusting and say how each works. 3

**17.** Explain why potassium and sodium react with water to produce flames. 3

OR

Define roasting and calcination. Give one equation for each. 3

**18.** What is meant by an alloy? State why an alloy is made. Give two alloys with their components. 3

OR

What are the reaction products when a metal oxide dissolves in water? Give an example with an equation. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Describe how metals are extracted according to their reactivity: low, medium and high reactive metals with an example each. 5

OR

Explain the extraction of aluminium or sodium by electrolysis. Why can they not be extracted by carbon? 5

**20.** Describe the electrolytic refining of copper. Name the anode, cathode and electrolyte and give the reactions. 5

OR

Explain the thermit reaction, the reaction of metal with non-metal and with dilute acid, with one equation each. 5

## END OF QUESTION PAPER — Total: 40 Marks

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